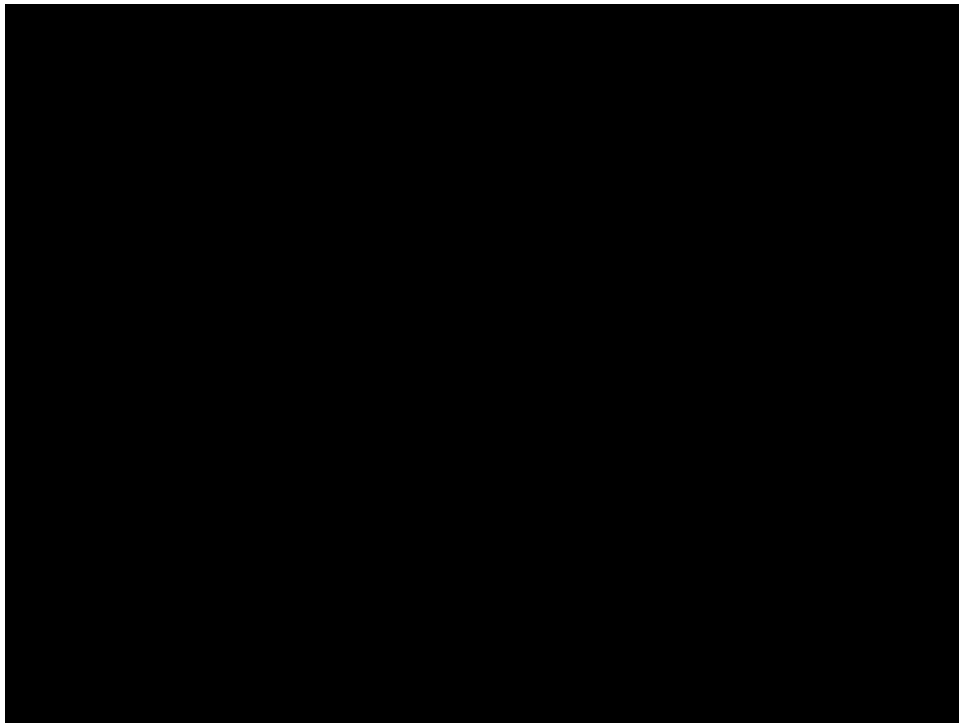

Video Sampling and Filtering

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IMCV- FEUP

Input sequences characteristics



- Dataset: YouTube 8M*
- Name: v_Biking_g14_c04.avi
- Frames: 156
- Dimensions: (240, 320, 3)
- Time: 0:00:05
- Frames per second: 29.97

* Source: <https://research.google.com/youtube8m/>

FIR

Objective:

Develop and compare two digital filters: one for temporal smoothing and one for temporal differentiating.

Specifications:

Temporal Smoothing Filter - Simple Average Filter

- Number of coefficients: 5
- Number of operations per pixel: 1
- Total Computation time: 0.398 seconds
- Computation time per frame: 0.002551282 seconds

Temporal Differentiating Filter - Central Differences Filter

- Number of coefficients: 3
- Number of operations per pixel: 4
- Computation time: 0.716 seconds
- Computation time per frame: 0.004589744 seconds

Total Computation Time = 1.113 seconds

FIR - Algorithms

Temporal Smoothing

Simple Averaging Filter

Number of Coefficients: 5

For each frame in the sequence:

1. Apply a weighted filter to each channel of the frame by taking the mean of the previous 5 frames.

2. Apply the

		Sample points				
Gain		-2	-1	0	1	2
1/3			1	1	1	
1/5		1	1	1	1	1

frame to
e.

		Sample points				
Gain		-2	-1	0	1	2
1/2			-1	0	1	
1/12		1	-8	0	8	-1

Temporal Differentiating Algorithm

Central Differences Filter

For each frame in the image sequence:

1. Compute central differences along the x and y axes using the Sobel operator.
2. Calculate the magnitude of the differences to obtain the central differences.
3. The temporal differentiation is performed by subtracting the gradient magnitudes of the current frame from the gradient magnitudes of a previous frame.
4. Append the resulting frame to the differentiated sequence.

$$\frac{\partial^{(1)} f}{\partial x^{(1)}} \approx \frac{-1f(x-1h) + 0f(x+0h) + 1f(x+1h)}{2h^1}$$

IIR Recursive

Objective:

Develop and compare two digital filters: one for temporal smoothing and one for temporal differentiating.

Specifications:

Temporal Smoothing Filter:

- Number of coefficients: 10 (2nd order)
- Number of operations per pixel: 8

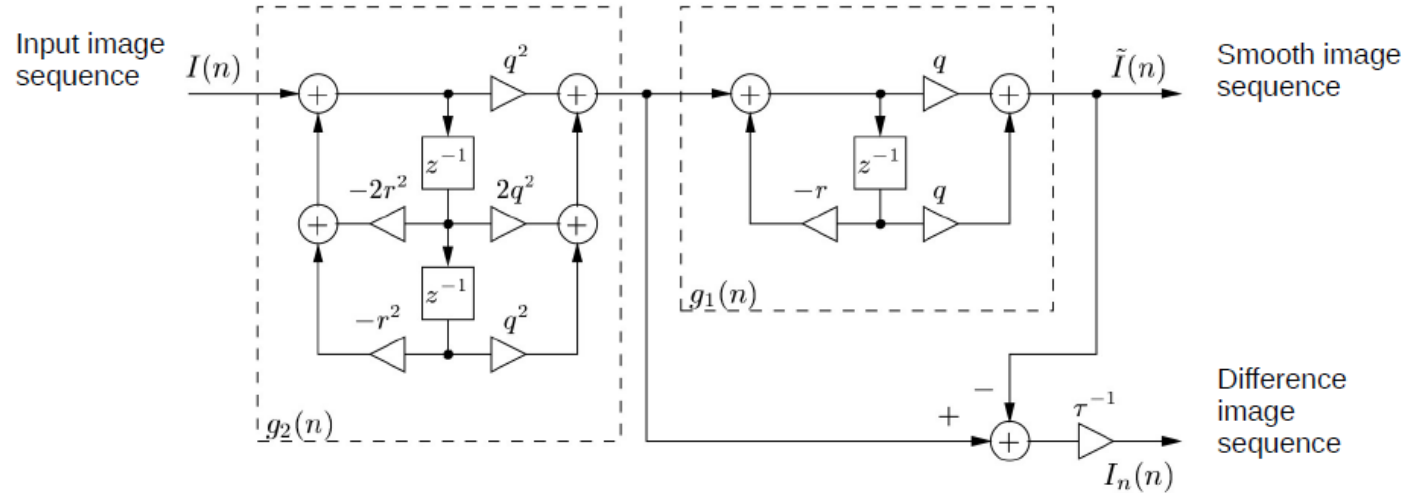
Temporal Differentiating Filter:

- Number of coefficients: 12 (2nd order)
- Number of operations per pixel: 9

Computation time per frame (both filters): 0.0303 seconds

Total computation time (both filters): 1.93 seconds

IIR Recursive - Algorithm



$$G_2(z) = q^2 \left(\frac{1 + 2z^{-1} + z^{-2}}{1 + 2rz^{-1} + r^2z^{-2}} \right); G_1(z) = q \left(\frac{1 + z^{-1}}{1 + rz^{-1}} \right); q = \frac{1}{1 + 2\tau}; r = \frac{1 - 2\tau}{1 + 2\tau}$$

Results: Metrics

	FIR		IIR	
	Smoothed	Differenced	Smoothed	Differenced
Contrast (%)	100.0	100.0	100.0	100.0
Sharpness (%)	94.18	100.0	96.79	100.0
PSNR (db)	25.14	7.95	29.96	7.92
SSIM (%)	93.90	31.67	99.0	32.0

Results: Output sequences

FIR

Smoothed



Differenced



IIR

Smoothed



Differenced



Thank you!